

Activity

Group members:

Dengue data

Here we will return to the dengue data from class. Our model is

$$Dengue_i \sim \text{Bernoulli}(\pi_i)$$
$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 WBC_i$$

Here is the output for the fitted model from R:

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.73743	0.08499	20.44	<2e-16 ***
WBC	-0.36085	0.01243	-29.03	<2e-16 ***

1. Calculate a 99% confidence interval for the change in the log odds of dengue associated with a one-unit increase in WBC.
2. How could we calculate a 99% confidence interval for the change in the *odds* of dengue associated with a one-unit increase in WBC?
3. Your friend, seeing your confidence interval from question 2, states that there is a 1% chance that the true change in odds associated with a one-unit increase in WBC is less than 0.675 or greater than 0.720. Is your friend correct?